Gate diagnosis: why real-world document recall is 20 points below the held-in estimate

Results

Arm B's document gate scores **0.8564** document recall on held-in real-world calibration and **0.6495–0.6790** on the sealed real-world corpora, while on synthetic data the same gate moves 0.9348 → 0.9311 — no gap at all. Three hypotheses were tested. **All three are refuted.** The cause is a property of the data, and it is deliberate.

What was ruled out

#	Hypothesis	Test	Verdict
a	Gate weights overfit (`alpha` 7.2e-7 over 262,144 features)	ROC-AUC held-in vs sealed, `alpha` swept 7.2e-7 → 1e-2	**refuted** — ΔAUC moves only –0.113 → –0.074 across four orders of magnitude
b	The threshold does not transfer	recall at the held-in cut vs at a sealed cut of matched specificity	**refuted** — re-cutting on sealed is *worse* (0.4979 vs 0.6504); at matched specificity the gap is larger, not smaller
c	The gold shifted between halves	judge fields, provenance mix, entity composition	**refuted** — same judges (gemini + opus), 63.7% vs 63.3% dual-judge agreement, 3.25 entities per positive in both

Two further candidates raised during the investigation, also refuted:

Candidate	Test	Verdict
Source-directory leakage across the held-in carve	carve whole directories instead of documents	**refuted** — held-in AUC 0.9894 → 0.9864. Barely moves
Non-random train/eval directory assignment	directory-number structure	**refuted** — 198 eval directories form 162 interleaved runs across 0–999, longest run 4

The decisive test was leave-one-corpus-out. A gate fitted with **govdocs2 removed from training entirely** still scores:

gate fitted on	govdocs2 held-in AUC	govdocs2 sealed AUC	Δ
all corpora	0.9894	0.8434	–0.1460
without govdocs2	0.9879	0.8378	–0.1501
without any real corpus	0.9865	0.8454	–0.1411

A model that has never seen a single govdocs2 document separates one half at 0.99 and the other at 0.84. Nothing about training can explain that.

What it actually is

The two halves are not equally separable, and both classes move toward each other. Scored by a gate trained on **synthetic data only**:

	n	p10	p25	p50	p75	p90
train positives	1,645	292	462	**759**	1,163	1,609
eval positives	3,484	–352	43	**520**	1,085	1,567
train negatives	1,849	–1,285	–957	–**609**	–314	–111
eval negatives	3,065	–1,052	–725	–**362**	–39	313

Class separation collapses from 1,368 to 882 — a 36% reduction. The eval half's positives carry subtler evidence *and* its negatives look more like PII. The 90th percentile of eval negatives (313) sits above the 10th percentile of eval positives (–352) and near their 25th (43).

Everything else matches: 100% `.txt` in both, identical length and feature-density distributions, and positive entity composition identical to within a ratio of 0.78–1.06 across all sixteen common entity types (`full name` 91.7% vs 91.7%, `phone number` 45.2% vs 45.0%, `email` 36.4% vs 37.6%).

The eval half is adversarial by construction, which is what the suite already says of it: `role: adversarial`, "the nearest thing in the suite to the customer's own corpus — heterogeneous real files, mostly mundane, mostly clean."

TL;DR

- The 20-point gap is **not a defect and not fixable by training**. A gate that never saw govdocs2 reproduces it exactly.
- It is the difference between a **random** sample of real documents (the training half) and an **adversarially selected** one (the sealed half). Class separation is 36% narrower in the sealed half.
- **The sealed number is the honest one.** 0.6495 document recall is what this architecture delivers on hard real-world documents; 0.8564 is what the training-side real corpus flatters it into looking like.
- **One real improvement survives**: source-balancing the gate's loss (real documents are 7.8% of fit rows) plus `alpha = 1e-2` lifts sealed real AUC 0.8453 → 0.8804 and sealed datax recall 0.5714 → 0.6529. Worth adopting.
- **The grouped-split fix should not be adopted** — it costs complexity and buys 0.003 AUC.
- The search's held-in objective will stay optimistic on real-world data as long as the training-side real corpus is not adversarial. That is the thing to fix, and it is a **data** problem, not a split or regularisation problem.

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Project	`projects/pii-head-to-head-v1`
Run ID	H1 (follow-up to the head-to-head)
Scope	document gate only; the 58 tag heads are untouched
Outcome	three hypotheses refuted; cause identified as adversarial construction of the sealed half; one adoptable improvement

Artifacts

Path	What
`probe/gate_diagnosis.json`	15 gate fits: alpha × balance, AUC and recall, held-in vs sealed
`probe/grouped_split.json`	per-document vs grouped carve, per corpus
`training/h2h_gate_diag.py`	the (a)/(b)/(c) separation
`training/h2h_grouped_split.py`	the grouped carve and its refutation
`reports/26-08-25_head-to-head.md`	the head-to-head this follows from

Model scoreboard

Model	Dataset	F1@5	F1@3	F1@1	Macro F1	Docs/sec	ms/doc
`fusion-1k`	10360_betterdataai_ner_silver_eval_10.36k				0.208	1360.8	0.7
`fusion-1k`	10626_ai4privacy_pii_masking_eval_10.63k				0.424	1360.8	0.7
`fusion-1k`	20000_pii_holdout_20.00k				0.477	1360.8	0.7
`fusion-1k`	30000_pii2_eval_25.15k				0.413	1360.8	0.7
`fusion-1k`	38937_openpii_pii_eval_38.94k				0.510	1360.8	0.7
`fusion-1k`	4000_datax-dualjudge-evalset-1.32k					1360.8	0.7
`fusion-1k`	5617_nemotron_eval_5.36k					1360.8	0.7
`fusion-1k`	6589_govdocs2-dualjudge-eval20-3.53k					1360.8	0.7
`steady-cascade`	10360_betterdataai_ner_silver_eval_10.36k				0.266	314.1	3.2
`steady-cascade`	10626_ai4privacy_pii_masking_eval_10.63k				0.729	314.1	3.2
`steady-cascade`	20000_pii_holdout_20.00k				0.709	314.1	3.2
`steady-cascade`	30000_pii2_eval_25.15k				0.757	314.1	3.2
`steady-cascade`	38937_openpii_pii_eval_38.94k				0.657	314.1	3.2

Model	Dataset	F1@5	F1@3	F1@1	Macro F1	Docs/sec	ms/doc
steady-cascade	4000_datax-dualjudge-evalset-1.32k					314.1	3.2
steady-cascade	5617_nemotron_eval_5.36k					314.1	3.2
steady-cascade	6589_govdocs2-dualjudge-eval20-3.53k					314.1	3.2
fusion-12k	10360_betterdataai_ner_silver_eval_10.36k				0.208	270.3	3.7
fusion-12k	10626_ai4privacy_pii_masking_eval_10.63k				0.425	270.3	3.7
fusion-12k	20000_pii_holdout_20.00k				0.485	270.3	3.7
fusion-12k	30000_pii2_eval_25.15k				0.390	270.3	3.7
fusion-12k	38937_openpii_pii_eval_38.94k				0.511	270.3	3.7
fusion-12k	4000_datax-dualjudge-evalset-1.32k					270.3	3.7
fusion-12k	5617_nemotron_eval_5.36k					270.3	3.7
fusion-12k	6589_govdocs2-dualjudge-eval20-3.53k					270.3	3.7

Measured 2026-08-25 · n=10,360, n=10,626, n=20,000, n=30,000, n=38,937, n=4,000, n=5,617, n=6,589.